

AJAY KRISHNA DEVULAPALLY

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PROFESSIONAL SUMMARY

AI/ML-focused Computer Science graduate student and Research Assistant at Penn State with experience building simulation-driven ML systems, reinforcement-learning workflows, GenAI prototypes, and deployment-oriented AI pipelines. Skilled in Python, C++, PyTorch, TensorFlow, LLMs, model evaluation, technical documentation, Git-based development, Agile collaboration, and translating research concepts into usable engineering workflows.

EDUCATION

The Pennsylvania State University

M.S. in Computer Science, Thesis Track

University Park, PA

Aug 2024 – Aug 2026

CVR College of Engineering

B.Tech. in Computer Science and Engineering

Hyderabad, India

2019 – 2023

EXPERIENCE

Research Assistant

2025 – Present

Microsystems Design Lab, Penn State | SRC

University Park, PA

- Conduct SRC-supported research under **Prof. Vijaykrishnan Narayanan** on AI/ML-driven optimization for microsystems, simulation workflows, and combinatorial design problems.
- Designed Python/PyTorch pipelines integrating simulation, routing optimization, benchmarking, model evaluation, and metric-driven analysis across industrial-scale environments.
- Document model assumptions, experimental methodology, evaluation metrics, failure modes, and solution-quality trade-offs for research reports, presentations, and technical discussions.

AI Intern

Mar 2021 – Apr 2021

Smartknower

Bangalore, India

- Built ANN-based and transfer-learning image classification models, achieving 98.3% accuracy on MNIST.
- Developed preprocessing, validation, evaluation, and deployment workflows using Python, OpenCV, Pandas, and Streamlit.

RESEARCH

TWARLeT-2.5D – First Author

2026

AI/ML for Simulation-Driven Chiplet Placement

- Designed an end-to-end reinforcement-learning system for thermally aware 2.5D chiplet placement using imitation learning, PPO, MILP-based routing, and thermal simulation.
- Built automated evaluation workflows across industrial-scale benchmarks, reducing wirelength by up to 27% and achieving 58× faster time to solution than simulated annealing.

PROJECTS

Simulation AI Plugin Prototype | Python, PyTorch, LLMs, APIs, Git

2026

- Designed a prototype AI assistant for simulation workflows with structured inputs, prompt-engineered LLM outputs, experiment summaries, model recommendations, and reproducible technical documentation.
- Mapped architecture to modular API services, Git-based version control, request validation, logging, evaluation checks, and cloud microservice deployment concepts.

GenAI RAG Document Intelligence Prototype | Python, LLMs, Embeddings, Vector Search

2026

- Built an LLM-powered RAG prototype for technical document analysis using embedding search, context retrieval, structured extraction, prompt engineering, and answer-quality evaluation.
- Created reusable documentation templates for model behavior, system architecture, limitations, evaluation criteria, and deployment-readiness review.

AI-Driven Optimization using Oscillator-Based Ising Machines | Python, C++, Optimization

2026

- Building oscillator-based Ising-machine optimization workflows for graph combinatorial problems, with metric-driven evaluation of convergence behavior, stability, and solution quality.

SKILLS

Programming: Python, C++, C, Java, SQL, TypeScript

AI / ML: PyTorch, TensorFlow, Keras, Scikit-learn, Reinforcement Learning, Imitation Learning, GNNs, Transformers, Computer Vision, Classical ML

Generative AI: LLMs, RAG, embeddings, vector search, prompt engineering, structured outputs, AI agents, evaluation workflows

Simulation / Systems: Simulation-driven evaluation, hardware-aware AI, optimization workflows, benchmarking, model debugging, deployment-oriented prototyping

Software Engineering: Git, Agile/Scrum concepts, software development lifecycle, technical documentation, REST API concepts, cloud microservices concepts, CI/CD concepts

Tools: Linux, Visual Studio Code, NumPy, Pandas, OpenCV, Streamlit, Tableau concepts, ScaleSim

Certificates: Artificial Intelligence – Smartknower (2021), Database Programming with SQL – Oracle (2022)

RELEVANT COURSEWORK

Graduate: Machine Learning, Graph Mining, Operating Systems, Algorithm Analysis, Computer Vision II

B.Tech: Artificial Intelligence, Deep Learning, Data Structures, Algorithms, DBMS, Software Engineering, Web Technologies